

Towards a Benchmark for Cyber Threat Intelligence Ontology Matching

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Abstract

Cybersecurity ontologies provide structured representations of threats, vulnerabilities, and defensive measures, yet their independent development has produced considerable semantic heterogeneity. Domain-specific evaluation resources for cybersecurity ontology matching remain limited. This paper presents a foundational reference-alignment benchmark for evaluating ontology matching systems within the Cyber Threat Intelligence (CTI) domain. The benchmark comprises 49 manually validated native cross-ontology equivalence mappings across ten ontology-pair scenarios derived from eight CTI resources: MITRE ATT&CK, CAPEC, CWE, CVE, STIX, CVSS, CPE, and D3FEND. It distinguishes between syntactic correspondences supported by lexical similarity and semantic correspondences representing conceptual equivalence despite lexical divergence. This distinction supports diagnostic analysis of matcher performance on lexically similar and lexically divergent mappings. Three representative ontology matching systems were evaluated: LogMap, AgreementMakerLight, and BERTMap. Within the evaluated scenarios, the results show stronger recovery of correspondences supported by lexical similarity than of conceptually equivalent mappings expressed through divergent CTI terminology. Given the small semantic subset, this pattern is interpreted as preliminary evidence of a semantic recovery gap within the scope of the benchmark. The benchmark is released as a reusable scientific resource under a CC BY 4.0 licence and includes reference alignments, the mapping registry, and reproducibility scripts. Future work will extend the resource through multi-expert validation, inter-annotator agreement analysis, CTI-specific background knowledge, and expressive alignment relations.

Keywords

Ontology matching, Cybersecurity ontologies, Semantic interoperability, Cyber Threat Intelligence

1. Introduction

The contemporary cybersecurity landscape is defined by a level of scale and complexity that increasingly exceeds the capacity of manual defensive processes. As detailed in the ENISA Threat Landscape 2024 report [1], the rise in automated and AI-driven cyberattacks has widened the attack surface and increased the need for more sophisticated incident detection mechanisms. To manage this complexity, organisations are increasingly shifting away from broad, standardised defensive baselines, often described as “general readiness” paradigms, which rely on static risk management practices and uniform solutions such as vendor patches and employee training [2]. Instead, there is growing emphasis on CTI efforts aimed at understanding adversarial behaviour through tailored and actionable knowledge relating to threat actors, vulnerabilities, and attack techniques [2, 3]. The focus on CTI in this study is deliberate because CTI concerns the representation and exchange of operational knowledge about adversarial behaviours, vulnerabilities, attack patterns, indicators, and defensive actions.

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To provide this structured knowledge, organisations rely on CTI resources such as MITRE ATT&CK, CAPEC, CWE, CVE, STIX, CVSS, CPE, and D3FEND [3, 4]. However, because these resources were developed independently to support different organisational contexts spanning strategic, tactical, and operational levels, they exhibit considerable semantic heterogeneity [2, 3, 4]. Differences in terminology, modelling structure, and conceptual granularity create a fragmented knowledge ecosystem that complicates the integration of threat data across platforms and reduces the effectiveness of automated analysis. Ontologies have emerged as an important mechanism for formalising CTI concepts into machine-interpretable representations [3]. Nevertheless, semantic fragmentation persists at the data-model level because cybersecurity ontology resources may represent related concepts using different terminology and modelling structures [4, 3].

Ontology matching, or alignment, addresses this challenge by identifying semantic correspondences between entities across disparate ontologies [5, 6, 7]. Although ontology matching has reached a high level of maturity in established domains such as biomedicine and the Semantic Web [7], focused evaluation on cybersecurity and CTI ontology representations has only recently begun to receive greater attention [8]. Existing benchmark evaluations within the OAEI have historically been dominated by STEM and biomedical datasets [9]. More recently, Dilworth et al. [8] introduced defence and national-security ontology matching tasks for OAEI 2026, including cybersecurity tasks such as UCO-STIX and STIX-D3FEND. These developments reinforce the value of domain-specific evaluation resources for specialised terminology and heterogeneous modelling practices. As demonstrated by Kraus et al. [9] in the Digital Humanities domain, such resources can provide evaluation settings that are not fully represented in general-purpose benchmark tracks. Furthermore, researchers have noted that many existing benchmarks do not clearly distinguish between surface-level lexical similarity and genuine conceptual equivalence [10].

The work presented here is motivated by the recognition that raw threat data cannot become actionable intelligence without the ability to identify meaningful conceptual relationships across heterogeneous CTI knowledge structures [2]. Building reliable reference resources for large-scale or structurally complex ontologies is an important prerequisite for advancing domain-specific ontology matching [6]. However, as emphasised in the development of consensual OAEI datasets, the construction of high-quality reference alignments remains a labour-intensive process requiring manual validation [11]. In addition, standard evaluation initiatives often focus on ontology pairs containing accepted correspondences, which may not reflect the sparse equivalence relationships observed in some specialised-domain scenarios [12].

To address these challenges, this paper introduces a foundational reference-alignment benchmark for CTI ontology matching. The benchmark comprises manually validated reference alignments across ten ontology-pair scenarios derived from eight CTI resources [3, 4]. A distinguishing feature of this dataset is its formal separation of syntactic correspondences based on lexical similarity and semantic correspondences based on conceptual equivalence [10, 9]. This granular structure enables diagnostic evaluation of whether current matchers can move beyond surface-level similarity to recover conceptually equivalent mappings across divergent CTI terminology. Within the evaluated scenarios, the observed difference between syntactic and semantic recovery is treated as preliminary evidence of a semantic recovery gap within the scope of the benchmark. In practical CTI workflows, unreliable alignments may hinder incident correlation, threat graph integration, and analyst trust in automated intelligence-sharing pipelines, further motivating the need for a domain-specific benchmark. The primary contributions of this work are as follows:

1. A foundational, manually validated reference-alignment benchmark for CTI ontology matching that distinguishes syntactic correspondences based on lexical similarity from semantic correspondences based on conceptual equivalence [10, 9].
2. A comparative evaluation of representative logic-based (LogMap), hybrid (AgreementMakerLight), and embedding-based (BERTMap) ontology matching systems [13, 14, 15] to establish a reproducible performance baseline for the CTI domain across ten ontology-pair scenarios.
3. A diagnostic analysis of alignment outcomes across the evaluated CTI resources, comparing

syntactic and semantic recovery and interpreting matcher performance in the context of modelling density and structural heterogeneity.

The remainder of this paper is organised as follows. Section 2 reviews related work and gaps in CTI ontology matching evaluation. Section 3 details the methodology and reference-alignment construction. Section 4 reports results across ten alignment scenarios. Section 5 discusses the findings and outlines future work, while Section 6 concludes the paper.

2. Related Work

This section reviews recent methodological developments in ontology matching and examines their applicability to the cybersecurity domain. Drawing on survey literature and benchmark-oriented studies published between 2020 and 2026, the section identifies the research context motivating the manually validated reference-alignment benchmark proposed in this study.

2.1. Methodological Evolution: From Rule-Based to Predictive Paradigms

Ontology matching has evolved from similarity-based and structural techniques toward hybrid, embedding-based, and transformer-driven approaches [7, 16, 17, 18, 19, 20, 21, 22]. Figure 4 illustrates this progression, showing the transition from lexical and structural similarity methods toward representation-learning and neural ontology matching approaches, as summarised in Appendix A.

Early ontology matching systems relied primarily on lexical and structural similarity measures, comparing ontology entities through string-based metrics and graph-based heuristics [7, 17, 22]. Subsequent approaches combined multiple similarity signals and incorporated external background knowledge [18, 7]. Systems such as AgreementMakerLight integrate lexical, structural, and semantic similarity strategies together with logical consistency mechanisms [14]. More recent approaches use representation learning to encode ontology entities as vector representations capable of supporting semantic comparison [21, 22]. BERTMap, for example, applies transformer-based contextual representations to ontology matching [15]. Embedding- and transformer-based approaches may also introduce additional computational requirements associated with representation generation and model inference. These developments provide a range of matching paradigms for evaluating heterogeneous ontology representations, including those found in specialised domains such as cybersecurity.

2.2. Cybersecurity Interoperability and Benchmarking

Cybersecurity knowledge representation has received increasing attention through ontology and knowledge-graph initiatives. Sikos [23] surveys cybersecurity knowledge graphs and highlights the role of formal knowledge representation in integrating heterogeneous cyber knowledge. Related initiatives include the Unified Cyber Ontology (UCO), a community-developed ontology designed to support representation across the cybersecurity domain, and the National Security Ontology Foundry, which seeks to support interoperability across defence and national-security ontologies [8]. These developments demonstrate continuing activity in cybersecurity and defence semantic interoperability.

The availability of domain-specific ontology matching benchmarks, however, has historically been more limited. A longitudinal assessment of OAEI tracks from 2004 to 2025 shows that STEM and biomedical datasets account for approximately 58% of the analysed tracks, with no dedicated cybersecurity or CTI track identified during that period. Figure 1 summarises this historical distribution. Similar to benchmark development in Digital Humanities [9], this motivates evaluation resources designed around specialised terminology and heterogeneous modelling practices.

More recently, Dilworth et al. [8] introduced the DISO-OAEI track for defence and national-security ontology matching. Their work analyses more than 60 publicly available ontologies and introduces eight matching tasks, including cybersecurity scenarios such as UCO-STIX and STIX-D3FEND. The track includes consensus alignments derived from multiple ontology matching systems and a manually curated

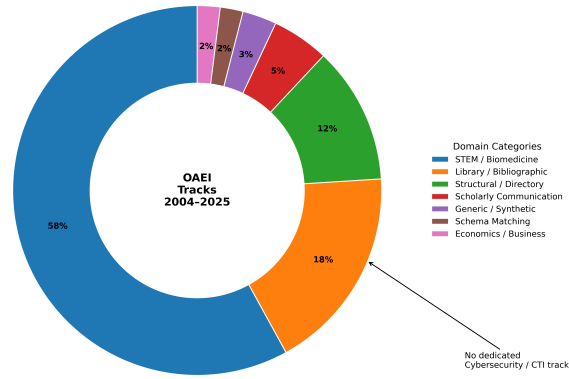


Figure 1: OAEI tracks by topic (2004–2025), showing STEM/biomedical dominance and no dedicated cybersecurity or CTI track.

silver-standard alignment constructed through validation of consensus and selected unique mappings, representing an important recent development in domain-specific ontology matching evaluation. The present benchmark complements this work by focusing specifically on CTI resources and manually validated native one-to-one equivalence correspondences, with an explicit diagnostic separation between syntactic and semantic reference mappings.

2.3. Challenges in Transferability and Complex Matching

Despite recent progress, applying general-purpose ontology matching methods to CTI remains challenging because cybersecurity resources frequently use specialised terminology, different levels of abstraction, and heterogeneous modelling structures. Appendix A reports a four-point qualitative assessment of ontology matching survey literature published between 2020 and 2025, based on the extent to which the surveyed studies discuss methods, resources, or challenges transferable to CTI ontology matching. The scores are descriptive rather than performance measures and indicate that recent methodological advances have potential relevance for cybersecurity, while their behaviour on CTI-specific ontology representations still requires domain-focused evaluation.

Hybrid ontology matching systems commonly use general-purpose background knowledge sources such as WordNet and DBpedia [18]. Such resources can support lexical and semantic matching but may provide limited coverage of specialised concepts involving adversarial tactics, attack procedures, vulnerabilities, threat actors, and defensive controls. Cybersecurity-specific ontologies and knowledge graphs may therefore provide useful complementary background knowledge [23]. Furthermore, complex correspondences, including broader, narrower, and many-to-many relationships between attack behaviours and defensive countermeasures, remain difficult to represent in evaluation settings focused primarily on one-to-one equivalence [7].

Within this context, the benchmark presented in this study provides a focused CTI evaluation resource across ten ontology-pair scenarios. Its purpose is not to replace broader cybersecurity or defence interoperability initiatives, but to provide a reproducible reference-alignment dataset for diagnosing matcher behaviour on syntactic and semantic CTI correspondences.

3. Methodology

The methodology integrates the construction of a foundational reference-alignment benchmark with a multi-paradigm evaluation design to assess the effectiveness of ontology matching systems within the CTI domain. As illustrated in Figure 2, the workflow transitions through four stages: resource selection, structural profiling, parallel evaluation tracks, and performance interpretation [9].

3.1. Stage 1: Resource Selection and Domain Requirements

To inform matcher selection and benchmark requirements, this study conducted a targeted review of ontology matching literature published between 2020 and 2025, with particular attention to methodological developments and their potential transferability to CTI [9].

The initial resource selection prioritised CTI resources that are publicly available, actively maintained, machine-processable, and widely used in cybersecurity research or operational threat intelligence. Candidate resources were considered based on ontology-like structure, explicit conceptual entities, cross-resource relevance, and suitability for native equivalence assessment.

- **Ontology Selection:** Eight CTI resources were selected based on industry adoption, structural diversity, and domain specificity: MITRE ATT&CK, CAPEC, CWE, CVE, STIX, CVSS, CPE, and D3FEND [3, 4].
- **System Selection:** Three systems were selected to represent different dominant matching paradigms: LogMap (logic-based) [13], AgreementMakerLight (AML) (hybrid) [14], and BERTMap (embedding-based) [15]. These labels describe the primary matching approach of each system, although the systems may also use additional lexical, structural, or reasoning components.

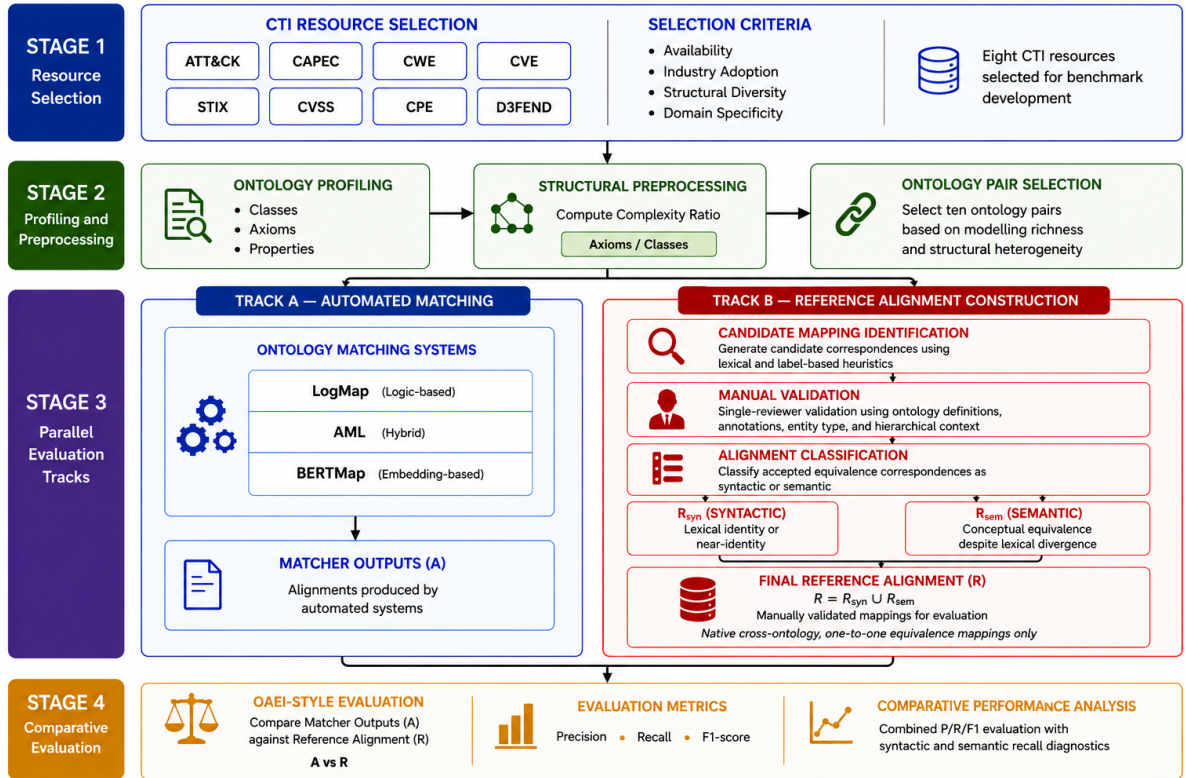


Figure 2: CTI benchmark workflow. Matcher outputs are evaluated against $R = R_{\text{syn}} \cup R_{\text{sem}}$, with syntactic and semantic subsets supporting diagnostic recall.

3.2. Stage 2: Profiling, Preprocessing, and Resource Cataloguing

The selected ontology representations were profiled in Protégé to extract structural metrics and organised into an Ontology Catalogue [3].

- **Complexity Ratio Calculation:** To provide a quantitative indicator of structural density, a Complexity Ratio (Logical Axioms/Class Count) was computed for each resource. This was the only derived quantitative structural metric used for comparison in this study. The analysis,

summarised in Table 1, provided structural context for interpreting the ten ontology-pair scenarios. The final set of pairs was selected to provide coverage across vulnerability, attack, defence, intelligence-sharing, platform, and risk-assessment resources while remaining feasible for manual validation and reference-alignment construction. The selected scenarios were not intended to exhaust all possible pairwise combinations.

- **Verification and Structural Integrity:** Each ontology was pre-tested in Protégé to verify successful parsing and basic structural validity. These checks were qualitative matcher-readiness checks rather than an additional numerical metric. This step confirmed that the ontology representations were compatible with the matching workflow and supporting libraries such as the OWL API and Apache Jena.

Table 1

Structural characteristics of CTI resources, with object and data properties reported as a combined Properties count.

Resource	Field / Role	Classes	Axioms	Properties	Complexity Ratio
CWE	Weaknesses	12	137	62	11.42
CAPEC	Attack Patterns	12	115	50	9.58
CPE	Platform Naming	5	32	16	6.40
CVSS	Risk Scoring	8	50	21	6.25
STIX	Intelligence Format	66	307	119	4.65
ATT&CK	Adversary Behaviour	14	64	25	4.57
D3FEND	Defensive Matrix	3495	12287	247	3.52
CVE	Vulnerabilities	10	26	11	2.60

3.3. Stage 3: Parallel Evaluation Tracks

The study implements two parallel tracks to maintain a clear distinction between the automated matching process and the manual construction of the reference alignments.

3.3.1. Track A: Automated Matching Experiments

Matching experiments were executed on a Windows 11 workstation equipped with an 11th Gen Intel(R) Core(TM) i7-11800H CPU (@ 2.30 GHz) and 32 GB RAM. LogMap v4.0 was executed using its standard command-line configuration, AML v3.2 using a fixed manual configuration (`config.ini`), and BERTMap through the DeepOnto v0.8.2 framework using a fixed experimental configuration.

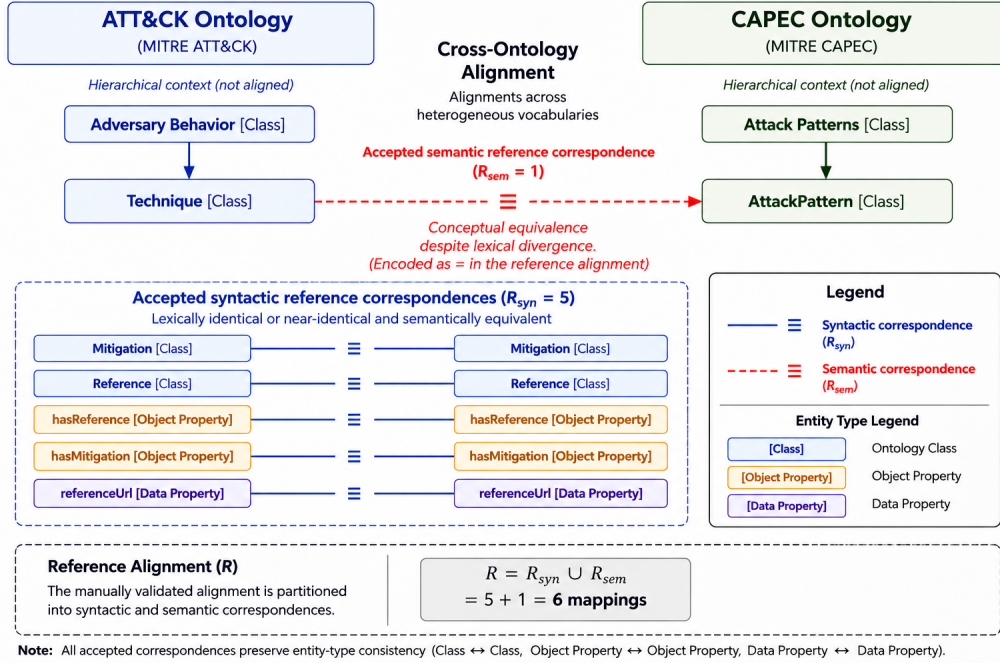
Each matcher ontology-pair experiment was executed three times. End-to-end wall-clock execution time was measured from matcher invocation to alignment generation, and the mean runtime was reported. Mapping counts were verified across repeated executions to confirm output consistency.

Reproducibility: A fixed directory structure was maintained for ontology inputs, runtime logs, raw matcher outputs, validated alignments, and runtime summaries. Configuration files, execution logs, and software version information were retained to support reproducibility. The public repository link for the released benchmark package is provided in Section 3.5.

3.3.2. Track B: Reference Alignment Construction (Human-in-the-Loop)

As illustrated in Figure. 2, a multi-stage validation protocol was implemented to establish the reference alignment (R) [9, 7]. Initial candidate correspondences were generated using lexical and label-based heuristics and then manually reviewed using Protégé and available ontology documentation. Candidate generation served as a starting point rather than as the sole source of accepted mappings; manual top-down inspection was additionally used to identify correspondences independently of lexical similarity. The outputs of the matchers evaluated in Track A were not used to determine the accepted reference correspondences. Figure. 3 shows ATT&CK–CAPEC as a representative reference-alignment example.

- **Manual Structural Sweep:** Manual validation used top-down inspection of ontology entities, definitions, annotations, entity types, and hierarchical context to identify lexical matches and non-lexical conceptual equivalences. The resulting benchmark was validated by a single researcher and provides a basis for future multi-expert refinement.
- **Term Match Definition:** Correspondences were considered valid only where both entities represented the same conceptual category and modelling role within the CTI context, rather than merely sharing lexical similarity. This criterion allowed lexically divergent equivalences while maintaining the benchmark’s strict equivalence scope.
- **Validation Guidelines:** Candidate correspondences were evaluated using formal definitions, annotations, entity type, and hierarchical context. Only high-confidence one-to-one equivalence mappings were retained. Ambiguous correspondences, broader or narrower semantic relations, imported metadata terms, and mappings requiring substantial interpretive assumptions were excluded from the reference alignment.
- **Classification:** Validated correspondences were partitioned into syntactic (R_{syn}) and semantic (R_{sem}) subsets. Syntactic correspondences represented lexical identity or near-identity, whereas semantic correspondences represented conceptual equivalence despite lexical divergence. These subsets were retained for diagnostic recall analysis.



Recall, and F1-score were calculated through exact correspondence matching under the benchmark’s one-to-one equivalence-only scope. For the embedding-based paradigm, BERTMap was executed through the DeepOnto framework. To adapt the synonym classifier to specialised CTI terminology, it was fine-tuned using the ATT&CK-CAPEC ontology pair before evaluation on the remaining ontology pairs. The ATT&CK-CAPEC pair was excluded from the BERTMap evaluation to avoid training-evaluation leakage. Diagnostic recall analysis was conducted using the syntactic and semantic subsets defined during reference-alignment construction. This enabled comparison of matcher recovery across lexically similar correspondences and conceptually equivalent but lexically divergent correspondences. Macro- and micro-averaged Precision, Recall, and F1-score were computed across ontology pairs with non-empty reference alignments. Pairs with empty reference alignments were treated as negative-control scenarios: matcher outputs on these pairs were false positives and were analysed separately.

Performance Interpretation: Quantitative results were interpreted alongside the Complexity Ratio and qualitative structural characteristics established in Stage 2. These characteristics provide contextual information for comparing heterogeneous CTI ontology pairs; the present study does not test a direct statistical relationship between structural complexity and matcher performance.

3.5. Data Availability

The benchmark package is publicly available under the CC BY 4.0 licence at <https://github.com/hkartey/cti-ontology-matching-benchmark>. It includes the ontology sources, reference alignments, mapping registry, evaluation outputs, runtime summaries, and reproducibility scripts.

4. Results

4.1. Characteristics of the Foundational CTI Reference Alignment

The constructed foundational CTI reference-alignment benchmark comprises ten ontology-pair scenarios derived from eight heterogeneous CTI resources. As detailed in Table 2, the benchmark distinguishes between syntactic correspondences (R_{syn}), based on lexical identity or near-identity, and semantic correspondences (R_{sem}), based on conceptual equivalence despite lexical divergence. The total reference set ($|R|$) contains 49 manually validated mappings, with syntactic correspondences accounting for 83.7% ($|R_{\text{syn}}| = 41$) and semantic correspondences accounting for 16.3% ($|R_{\text{sem}}| = 8$).

The number of accepted correspondences varies across the benchmark scenarios. CAPEC-CWE contains the largest reference set ($|R| = 32$), followed by ATT&CK-CAPEC ($|R| = 6$) and ATT&CK-STIX ($|R| = 5$). CAPEC-STIX contains a single semantic correspondence.

Four ontology pairs produced no accepted native one-to-one equivalence mappings: ATT&CK-D3FEND, CAPEC-D3FEND, CVE-CPE, and CVE-STIX. These pairs were retained as negative-control cases in the pair-level evaluation but excluded from aggregate Precision, Recall, and F1-score calculations because no accepted reference correspondences were available.

4.2. Aggregated Performance and Semantic Recovery

Table 3 presents pair-level evaluation results for ontology pairs with non-empty reference alignments. Macro-averaged Precision, Recall, and F1-score were used as the primary aggregate measures to assign equal weight to each evaluated ontology pair and avoid dominance by denser reference alignments such as CAPEC-CWE.

LogMap achieved the strongest overall performance, with a macro F1-score of 0.7153, followed by AML with 0.5973. LogMap achieved the highest macro Precision (0.7816), while AML achieved a macro Recall of 0.6951 compared with 0.7003 for LogMap. BERTMap produced fewer recovered reference correspondences, with a macro Recall of 0.0917 and macro F1-score of 0.1444.

BERTMap was not evaluated on the ATT&CK-CAPEC pair because this pair was used for synonym-classifier fine-tuning and was excluded to avoid training-evaluation leakage. The four ontology pairs

Table 2

Composition of the CTI reference alignment across the ten ontology-pair (O_1 - O_2) scenarios.

Scenario	O_1 Classes	O_2 Classes	R_{syn}	R_{sem}	$ R $
CAPEC-CWE	12	12	29	3	32
ATT&CK-CAPEC	14	12	5	1	6
ATT&CK-D3FEND	14	3495	0	0	0
CVE-CVSS	10	8	2	0	2
ATT&CK-STIX	14	66	2	3	5
CWE-CVE	12	10	3	0	3
CAPEC-STIX	12	66	0	1	1
CAPEC-D3FEND	12	3495	0	0	0
CVE-STIX	10	66	0	0	0
CVE-CPE	10	5	0	0	0
Total	–	–	41	8	49

with empty reference alignments were retained as negative-control cases in the pair-level analysis but excluded from macro-averaged Precision, Recall, and F1-score calculations.

The CAPEC-STIX scenario represents a semantic-only case: its single accepted correspondence, *CAPEC Mitigation = STIX CourseOfAction*, was not recovered by any of the evaluated matchers.

Table 3

Pair-level matcher performance against the combined reference alignment.

Ontology Pair	AML			LogMap			BERTMap		
	P	R	F1	P	R	F1	P	R	F1
ATT&CK-CAPEC	0.6250	0.8333	0.7143	1.0000	0.8333	0.9091	–	–	–
ATT&CK-STIX	0.2857	0.4000	0.3333	1.0000	0.4000	0.5714	0.0000	0.0000	0.0000
CAPEC-CWE	0.8108	0.9375	0.8696	0.9394	0.9688	0.9538	1.0000	0.1250	0.2222
CAPEC-STIX	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
CVE-CVSS	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.0000	0.0000	0.0000
CWE-CVE	0.5000	1.0000	0.6667	0.7500	1.0000	0.8571	1.0000	0.3333	0.5000
Macro Average	0.5369	0.6951	0.5973	0.7816	0.7003	0.7153	0.4000	0.0917	0.1444

4.3. Diagnostic Analysis of Syntactic and Semantic Recovery

The syntactic and semantic reference subsets were retained for diagnostic recall analysis. Subset-specific Precision and F1-score were not calculated because matcher outputs were not independently classified as syntactic or semantic predictions. Consequently, mappings belonging to one subset cannot be treated as false positives when evaluating recovery of the other subset. Table 4 presents aggregate syntactic and semantic recall across the evaluated ontology pairs. AML and LogMap recovered almost all syntactic reference correspondences, each achieving a syntactic macro recall of 0.9931. Semantic recovery was lower within the evaluated scenarios: AML achieved a semantic macro recall of 0.1667, LogMap achieved 0.2500, and fine-tuned BERTMap recovered no accepted semantic correspondences. Because the semantic subset contains only eight accepted correspondences, these results are interpreted as diagnostic evidence within the present benchmark rather than as a general estimate of CTI semantic matching performance.

To provide pair-level transparency, Table 5 reports semantic true positives and false negatives for ontology pairs containing accepted semantic correspondences. Overall in-scope false-positive counts are included only to contextualise matcher over-generation at the ontology-pair level. These false positives are not assigned to the semantic subset because matcher outputs do not distinguish syntactic from semantic predictions. Semantic Precision and F1-score are therefore not reported.

4.4. Negative-Control Cases

Four ontology pairs contained no accepted native one-to-one equivalence mappings: ATT&CK–D3FEND, CAPEC–D3FEND, CVE–CPE, and CVE–STIX. These cases were excluded from aggregate Precision, Recall, and F1-score calculations because Recall and F1-score are not meaningful when no accepted reference correspondences are available. Nevertheless, they were retained as negative-control cases to record in-scope equivalence mappings generated by each matcher. Because the reference alignment contains no accepted mappings for these pairs, all reported in-scope matcher outputs constitute false positives under the benchmark scope.

The absence of accepted equivalence mappings in the D3FEND scenarios should be interpreted in the context of the benchmark’s strict one-to-one equivalence scope. In the evaluated representations, D3FEND represents defensive techniques and countermeasures, whereas ATT&CK and CAPEC primarily represent adversarial behaviours and attack patterns. The absence of accepted ATT&CK–D3FEND and CAPEC–D3FEND equivalence mappings therefore does not imply an absence of meaningful CTI relationships between these resources. Many plausible cross-resource relationships are defensive, broader/narrower, or otherwise operationally related and fall outside the equivalence-only scope of the present benchmark.

Table 4

Diagnostic syntactic and semantic correspondence recovery. BERTMap totals exclude the ATT&CK–CAPEC pair used for synonym-classifier fine-tuning, leaving 36 syntactic and 7 semantic reference correspondences for BERTMap evaluation.

Matcher	Subset	Pairs	Recovered / Total	Macro Recall
AML	Syntactic	5	40 / 41	0.9931
AML	Semantic	4	2 / 8	0.1667
LogMap	Syntactic	5	40 / 41	0.9931
LogMap	Semantic	4	3 / 8	0.2500
BERTMap	Syntactic	4	5 / 36	0.1178
BERTMap	Semantic	3	0 / 7	0.0000

Table 5

Semantic correspondence recovery and overall false-positive counts.

Ontology Pair	$ R_{\text{sem}} $	AML			LogMap			BERTMap		
		TP	FN	FP*	TP	FN	FP*	TP	FN	FP*
ATT&CK–CAPEC	1	0	1	3	0	1	0	–	–	–
ATT&CK–STIX	3	0	3	5	0	3	0	0	3	0
CAPEC–CWE	3	2	1	7	3	0	2	0	3	0
CAPEC–STIX	1	0	1	8	0	1	0	0	1	0
Total	8	2	6	23	3	5	2	0	7	0

*FP denotes all unmatched in-scope predictions for the ontology pair, not semantic-subset false positives.

Table 6

In-scope mappings generated for empty-reference negative-control cases.

Ontology Pair	AML	LogMap	BERTMap
ATT&CK–D3FEND	9	1	5
CAPEC–D3FEND	6	3	0
CVE–CPE	0	0	0
CVE–STIX	4	0	0

5. Discussion

This section interprets the methodological and structural factors associated with CTI ontology matching performance and outlines the boundaries of the present benchmark.

5.1. Methodological Factors and the Semantic Recovery Gap

AML and LogMap recovered almost all syntactic reference mappings, each achieving a syntactic macro recall of 0.9931. Semantic macro recall was lower, reaching 0.1667 for AML and 0.2500 for LogMap, while fine-tuned BERTMap recovered no accepted semantic correspondences.

Within the evaluated scenarios, these results provide preliminary evidence of a semantic recovery gap between lexically similar and lexically divergent equivalence mappings. The evaluated systems performed strongly when equivalence was supported by lexical similarity but recovered fewer mappings where equivalence depended on specialised CTI terminology or ontology context. Because the semantic subset contains only eight accepted correspondences, this pattern should be interpreted as diagnostic evidence within the present benchmark rather than as a general conclusion about CTI ontology matching.

Several missed correspondences illustrate this pattern. No matcher recovered the accepted ATT&CK *Technique*–STIX *AttackPattern* correspondence or the CAPEC *Mitigation*–STIX *CourseOfAction* correspondence. The ATT&CK *AdversaryGroup*–STIX *ThreatActor* correspondence was also not recovered, despite the conceptual relationship represented in the reference alignment.

BERTMap and Background-Knowledge Limitations. BERTMap’s low semantic recovery should be interpreted in light of the limited fine-tuning setup, since its synonym classifier was fine-tuned using only the ATT&CK–CAPEC pair. The result therefore does not establish a general limitation of embedding-based ontology matching. Similarly, AML’s use of external background knowledge [18], including UBERON, did not recover several CTI-specific semantic correspondences, suggesting that domain-specific training data and cybersecurity knowledge resources may provide more relevant support.

5.2. Structural and Domain-Specific Challenges

The observed semantic recovery gap should also be interpreted in the context of differences in terminology, modelling objectives, and abstraction levels across independently developed CTI resources.

- **Structural Context.** Structural profiling was included because ontology matching systems may benefit from class hierarchies, logical axioms, and property structures. Although the present study does not test a direct statistical relationship between Complexity Ratio and matcher performance, these characteristics provide useful context when comparing heterogeneous ontology pairs.
- **Abstraction Levels and Divergence.** CTI resources differ in their modelling objectives and levels of abstraction, which can give rise to broader, narrower, inverse, defensive, and one-to-many relationships that fall outside the benchmark’s strict one-to-one equivalence scope. This is particularly relevant to the D3FEND scenarios, where the absence of accepted equivalence mappings does not imply an absence of meaningful cross-resource relationships.

5.3. Limitations

The benchmark and evaluation have the following limitations:

- **Reference Alignment Size and Scope.** The reference alignment contains 49 mappings, including only eight semantic correspondences. The semantic-recovery results should therefore be regarded as diagnostic within the evaluated scenarios rather than as a general estimate of CTI ontology matching performance.

- **Single-Researcher Validation.** Candidate mappings were identified through lexical heuristics and manual inspection, with final validation conducted by a single researcher. Although the mapping protocol was applied consistently, valid correspondences may have been missed and no inter-annotator agreement was measured.
- **Equivalence and Pair Coverage.** The benchmark is restricted to native one-to-one equivalence mappings and ten selected ontology-pair scenarios. It therefore excludes broader, narrower, inverse, related, and one-to-many relationships, as well as other potentially relevant pair combinations.
- **Absence of Downstream Evaluation.** Matcher performance was assessed using ontology-matching metrics rather than downstream CTI tasks such as incident correlation, SOC analyst support, or threat intelligence integration.

5.4. Future Work

Future benchmark development will focus on the following directions:

- **Multi-Expert Validation.** Future versions will introduce multi-expert review, adjudication, and inter-annotator agreement analysis to strengthen validation of ambiguous semantic correspondences.
- **Expanded Coverage and Expressive Alignments.** The benchmark will be extended with a larger semantic reference subset, additional ontology-pair scenarios, including STIX-D3FEND, and relations beyond one-to-one equivalence, such as broader, narrower, and $n : m$ correspondences.
- **CTI-Specific Matching Resources.** Future evaluation will investigate cybersecurity-specific background knowledge, mediator ontologies, knowledge graphs, and domain-adapted language models such as SecureBERT [24] to improve the recovery of lexically divergent CTI correspondences.
- **Benchmark Evolution and Emerging Approaches.** Further work will compare established matchers with neuro-symbolic, LLM-based, and retrieval-augmented approaches, while supporting versioned benchmark maintenance, reproducible evaluation, and future OAEI-style benchmarking initiatives.

6. Conclusion

This study presents a foundational CTI reference-alignment benchmark comprising eight CTI resources, ten ontology-pair scenarios, and 49 manually validated native cross-ontology equivalence mappings. The reference alignment distinguishes 41 syntactic and eight semantic correspondences, supporting diagnostic evaluation of lexically similar and lexically divergent mappings.

Across the evaluated scenarios, LogMap achieved the strongest macro F1-score (0.7153), followed by AML (0.5973) and fine-tuned BERTMap (0.1444). AML and LogMap recovered almost all syntactic correspondences, while semantic recovery was lower. Given the small semantic subset, this difference provides preliminary evidence of a semantic recovery gap within the present benchmark rather than a general conclusion about CTI ontology matching.

The released benchmark provides reference alignments, a validated mapping registry, evaluation outputs, and reproducibility scripts for future CTI ontology-matching research. Future extensions will focus on multi-expert validation, broader semantic and ontology-pair coverage, CTI-specific matching resources, and more expressive alignment relations.

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Declaration on Generative AI

During the preparation of this work, the authors used ChatGPT in order to: Assist with grammar check and the aesthetic design and visual layout of technical diagrams (Figures 2, 3, and 4). After using these services, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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A. Supplementary Methodological Figures

Figure 4 provides supplementary context for the methodological overview discussed in Section 2.1. Figure 5 provides supplementary context for the discussion in Section 2.3.

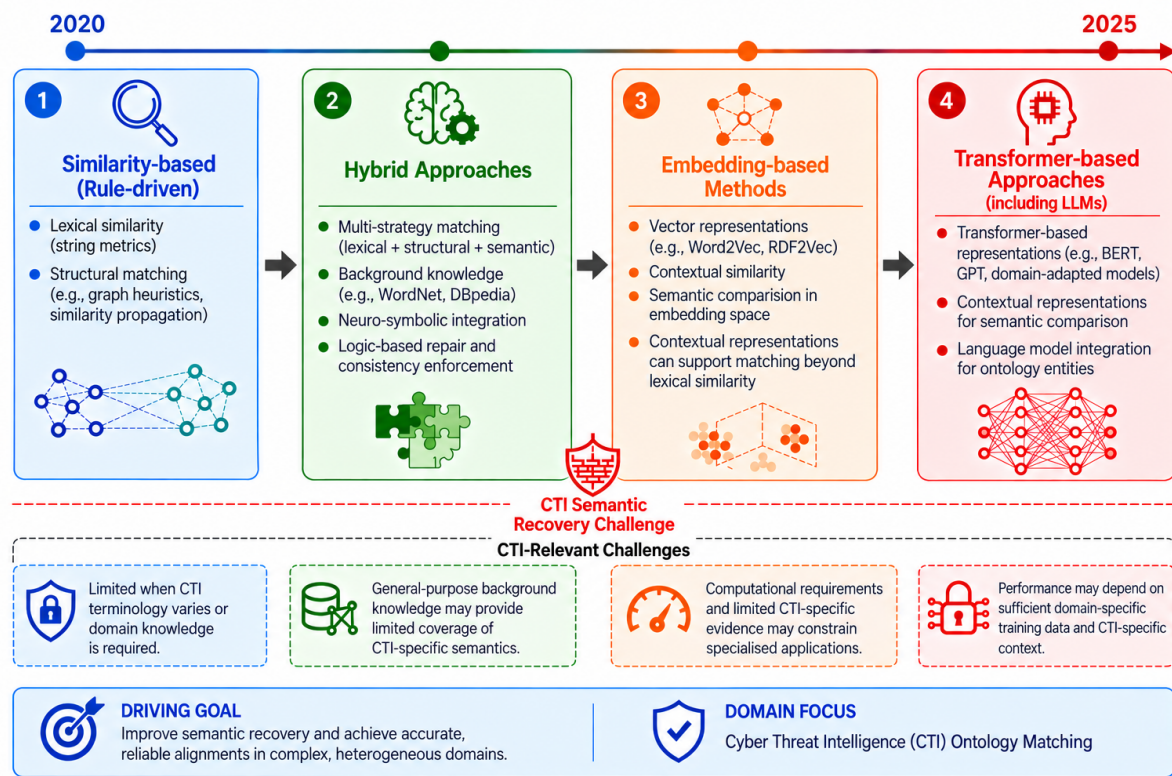


Figure 4: Schematic progression of representative ontology matching approaches identified in survey literature published between 2020 and 2025, highlighting overlapping methodological developments and challenges relevant to CTI ontology matching.

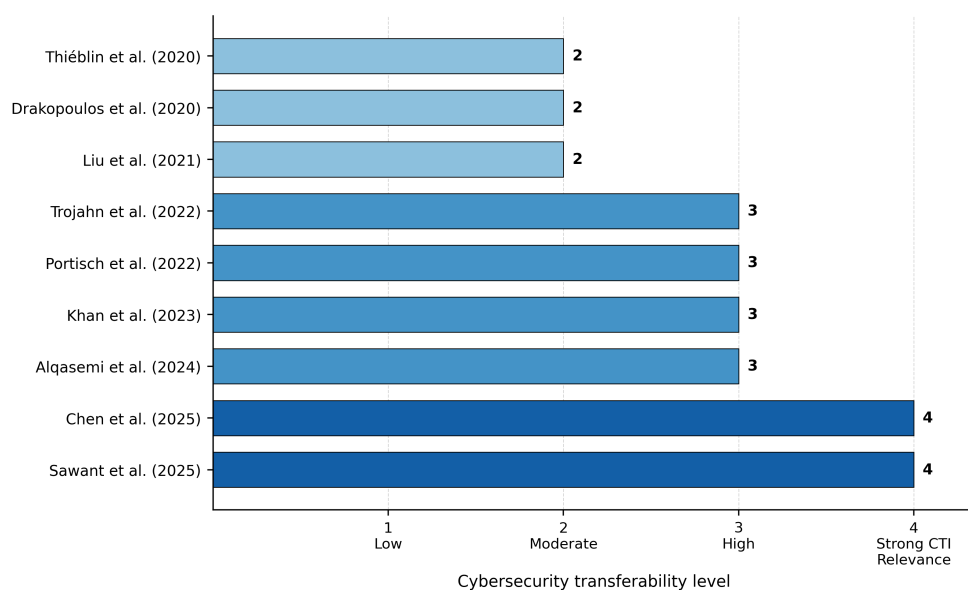


Figure 5: Qualitative assessment of CTI transferability in ontology matching survey literature published between 2020 and 2025.

The cybersecurity relevance assessment in Figure 5 was constructed using a four-point qualitative ordinal coding scheme to assess the transferability of recent ontology matching survey literature to CTI ontology alignment. The score is not a performance metric; it indicates the extent to which each survey discusses methods, resources, or challenges applicable to cybersecurity ontology matching. Scores of 1, 2, 3, and 4 represent low, moderate, high, and strong CTI transferability, respectively.

B. Ontology Provenance

Version and provenance metadata for the evaluated ontology files are reported in Table 7.

Table 7: Ontology provenance and structural statistics for the CTI resources used in the benchmark.

Ontology	Language	Metadata Source	Version / Release	Access Date	Classes	Axioms	Properties	Complexity Ratio
ATT&CK	OWL/RDF	SEPSES	2.0.0	20-Nov-25	14	64	25	4.57
CAPEC	OWL/RDF	SEPSES	2.0.0	10-Nov-25	12	115	50	9.58
CWE	OWL/RDF	SEPSES	2.0.0	10-Nov-25	12	137	62	11.42
STIX	OWL/RDF	VISTology	Not specified in OWL metadata	10-Nov-25	66	307	119	4.65
D3FEND	OWL/RDF	MITRE	1.2.0; 2025-08-01	10-Nov-25	3495	12287	247	3.52
CVE	OWL/RDF	SEPSES	2.0.0	10-Nov-25	10	26	11	2.60
CVSS	OWL/RDF	SEPSES	2.0.0	10-Nov-25	8	50	21	6.25
CPE	OWL/RDF	SEPSES	1.0.0	10-Nov-25	5	32	16	6.40

Note: Version information was extracted from OWL metadata where available. STIX contained no explicit version statement in the OWL metadata. Source URLs are provided in the project repository. Access date refers to the date on which the ontology file was obtained. Structural statistics correspond to the evaluated ontology files.

C. Runtime and Reproducibility Results

Each matcher was executed three times for each ontology pair. Runtime values are reported as the mean execution time across three runs; standard deviation (SD) indicates variation between repeated executions. Mapping reproducibility was assessed by comparing the generated alignment hashes and mapping counts across runs.

Table 8: Mean runtime and output reproducibility across ontology-pair scenarios.

Ontology Pair	AML		LogMap		BERTMap	
	Mean (s)	SD	Mean (s)	SD	Mean (s)	SD
ATT&CK-CAPEC	5.450	0.455	1.654	0.008	—	—
ATT&CK-D3FEND	6.838	0.167	6.096	0.157	6.721	1.576
ATT&CK-STIX	5.831	0.078	1.887	0.020	1.460	0.027
CAPEC-CWE	6.087	0.040	1.596	0.018	1.413	0.086
CAPEC-D3FEND	7.439	0.096	6.568	0.215	5.649	0.220
CAPEC-STIX	6.337	0.123	1.848	0.025	1.611	0.032
CVE-CPE	5.800	0.023	1.669	0.034	1.333	0.054
CVE-CVSS	6.005	0.063	1.697	0.022	1.467	0.065
CVE-STIX	6.185	0.166	1.991	0.015	1.599	0.058
CWE-CVE	6.033	0.148	1.728	0.029	1.584	0.071
Mean per completed pair	6.201	—	2.673	—	2.537	—
Total across completed pairs	62.005	—	26.734	—	22.837	—

Note: Alignment hashes and mapping counts were identical across repeated runs for all completed scenarios. BERTMap was not evaluated on ATT&CK-CAPEC because that pair was used for fine-tuning.